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Rapid urbanization and uneven development of the food retail environment

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ABSTRACT

We investigate how the development of supermarkets and public markets in a large, rapidly urbanizing African city can shape the equitability of the food system. Using remotely sensed retailer location and population datasets, we analyze the spatial-temporal development of supermarkets and public markets in Lusaka, Zambia from 2004 to 2020. As Lusaka's population grew by 116%, we found that supermarkets increased from 9 to 54 (500%) while public markets increased from 52 to 65 (25%). Using SaTScan spatial-temporal analysis, we identified seven significant areas of high supermarket development ($p < 0.05$) clustered from 2013 to 2020 in lower density, higher income residential areas. During this period ten new community public markets were established, suggesting a lack of direct involvement by the government to establish public markets at a pace similar to population growth. Supermarket development without concomitant development of public markets can negatively impact food system equitability in two ways: 1) supermarket development is spatially uneven, favoring more food secure residential areas, thus limiting food access; 2) supermarket development is outpacing the establishment of public markets, which have more favorable food purchasing conditions for low-income households. Improving food security in rapidly growing Southern African cities will require policies that promote retailer diversity and aim to improve urban food system resilience and equitability.

1. Introduction

High rates of poverty and food insecurity are co-occurring in urban Southern Africa (FAO 2021; World Bank 2018). At the same time, high urbanization rates have led to rapidly growing urban areas (Zimmer et al. 2020), both in terms of population growth and the outward expansion of urban spatial extents (d'Amour et al., 2017; United Nations, 2019). When urban growth rates outpace the ability of city planners to account for this rapid growth, high density unplanned settlements can develop (Cobbinah et al., 2015). Urban population and spatial growth without concomitant economic growth or city planning can contribute to increased rates of urban poverty and food insecurity, and exacerbate existing urban inequalities including those related to health, education, sanitation, and food systems (Cobbinah et al. 2015; Frayne & McCordic 2015; Hove et al. 2013).

Food systems are made up of producers, processors, distributors (i.e., retailers), and consumers (Ericksen, 2008). As the populations and

spatial extents of cities expand, the need to ensure sufficient, stable, sustainable, and equitable food access is necessary to achieve food security. In urban Southern Africa, household food access primarily occurs through purchasing from food retailers, including street vendors, small shops, wholesale outlets, public markets, and supermarkets, among others. Non-retail food access strategies can support food security among urban households, for example food sharing among socially connected households and urban agriculture (Battersby 2011; Miccoli et al., 2016), but most people purchase food from retailers (Davies et al. 2020). Understanding the immediate food retail environment of a household and how it is changing is critical for gaining insight into long standing food system inequalities that can hamper food security.

The food retail environment in Southern Africa is transforming, in part due to the implementation of policies that privilege and promote the growth of the supermarket sector (das Nair, 2018). Supermarket expansion in developing countries is often framed as a seemingly inevitable component of agricultural value chain development (see Barrett

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et al. 2020). Yet, the development of agricultural value chains may not be even, which can increase food system inequalities and vulnerabilities (Moseley and Battersby 2020). It is important to understand the spatial-temporal development trends of urban food retail environments, as these trends shape the food-related behaviors that households adopt. In short, the spatial and structural inequalities within the food system and broader urban context shape food access and, ultimately, food security (Smith et al. 2022; Shannon 2016; Frayne and McCordic 2015). Assessing food system equitability requires an integration of spatial and temporal characteristics to understand food access beyond assessments conducted at a single point in time (Widener and Shannon, 2014). Understanding how the food retail environment is changing in the context of urbanization and spatial expansion is important for observing how the development of different food retailers contributes to food access across space and time.

In this paper, we investigate the development of the food retail environment in Lusaka, Zambia, from 2004 to 2020, focusing on the spatial-temporal development of both supermarkets and public markets. Supermarkets are large, privately-owned stores that sell food products and non-food commodities (Cambridge 2023), while public markets are centralized sites which may or may not have permanent structures from where small-scale traders sell food products, non-food commodities, and / or provide services (Dewar & Watson 2018). We focus on the spatial-temporal development of supermarkets and public markets for three reasons. First, the growth of the supermarket sector in low- and middle-income countries is contributing to urban food system transformations (Berger and van Helvoirt, 2018), whereas public markets have long served as the centerpiece of food distribution - either by selling directly to consumers, acting as wholesalers to other retailers, or a combination of the two. The effectiveness and importance of public markets as sites of food distribution can depend on their governance structures, including their relationship with local governments (Davies et al. 2022). Second, a complex relationship exists among different food retailers within local food systems, and it is unclear how these roles change over space and time. Third, there emerge food system equity implications from the growth of the supermarket sector because, in the Zambian context, supermarkets are most often used by middle- and high-income households - not the poor who are most likely to be food insecure. This research contributes to a notable gap in the literature as much of the previous scholarly work on urban food policy development has emphasized food production and consumption (Haysom 2021; Khonje & Qaim 2019), yet the role of policy in shaping equitable urban food distribution systems is equally important. The supermarket sector is rapidly growing in Africa, and it is unclear how this growth trend will influence urban food retail environments and food systems. Understanding how urban food retailer environments develop over time and space is an important step for ensuring food security for urban households across the socio-economic spectrum. We address the following research questions:

1. How has the rate of supermarket, community public market, and government public market expansion changed over space and time in an expanding large Southern African city?
2. Are there differences in the spatial distribution and temporal development of public markets and supermarkets across space and time?

We first provide an overview of urban food retail environments in Southern Africa, detailing the roles of supermarkets and public markets as actors within urban food systems throughout the region, and describe how consumer access to these retailer types is linked to food consumption choices and household food security. In Section 3, we describe the study site of Lusaka, Zambia, and the remotely sensed data sets that we compiled to map and analyze the spatial-temporal trends of supermarket and public market distribution and development from 2004 to 2020. In Section 4 we describe how we used SaTScan spatial-temporal analysis to identify statistically significant areas of low and high supermarket

development across Lusaka from 2004 and 2020. Previously, Battersby (2017) assessed the spatial-temporal development of supermarkets over a 20-year period in Cape Town, South Africa, showing that supermarkets have rapidly expanded in the city, including into lower-income areas through the development of smaller supermarket retail outlets. However, to our knowledge our study is the first to comparatively assess the spatial-temporal development of supermarkets and public markets in the African context using remotely sensed data. We present the results of our mapping and statistical analysis in Section 5. In Section 6, we discuss our results in the context of equity within the food retail environment.

2. Urban food retailers in Southern Africa

Urban populations in Southern Africa are rapidly increasing and are expected to grow by more than 125 million people, from about 79 million in 2015 to 204 million people by 2050 (United Nations 2019). At the same time that urban populations are growing, the spatial extents of cities in Southern Africa are also expanding - typically through a “checkerboard” form of land development in which the layout of residential areas, infrastructure, and basic services can be spatially uneven and disconnected (Seto & Ramankutty, 2016). This form of urban growth can exacerbate existing social and economic inequalities (Hemerijckx et al., 2020), because higher income communities enjoy greater infrastructural investments than lower-income areas. The built environment and accessible infrastructure (e.g., transportation, food retailers, and potable water) of an area contributes to the ways in which urban residents access food (Frayne & McCordic 2015; Alkon et al. 2013).

Food systems are usually not explicitly considered by urban planners in Southern Africa (Haysom 2021), even though urban planning impacts food systems in various ways. For example, land use zoning policies may guide where public markets are established, or incentivize the development of shopping malls, many of which include supermarkets as anchor stores (Battersby, 2017). The spatial distribution and diversity of food retailers is critical for equitable food access in urban Southern Africa because consumers will choose where to purchase food based on their needs and available resources. For example, food retailers operate at different locations within urban areas, source food through diverse supply chains, offer different quantities of food at variable prices and levels of quality, and offer different amenities such as credit and longer operating hours (Wanyama et al. 2019; Tuholske et al. 2020; Battersby and Watson 2018). Consumer food access in urban Southern Africa is a complex web of retailer interactions, rather than a repeat relationship with a single retail outlet. The type, location, and development of retailers over time is critical to household food access and therefore food security outcomes. Thus, while food insecurity is experienced at the household scale, it is often the outcome of broader food system dynamics (Ericksen, 2008).

Urban food systems in Southern Africa are transforming through integration of global food supply chains into local food systems, in part, through the growth of the supermarket sector (Reardon et al., 2021). While supermarkets are becoming increasingly important actors in the food system, public markets - many of which predate colonial independence (Monteith, 2015) - remain important actors as well. Supermarkets and public markets are important for urban food retail environments for three reasons: 1) both market types serve as points of entry into the city for domestically and internationally produced food; 2) both provide food directly to consumers; and 3) both are used as wholesalers by other food retailers (i.e., small shops, other public markets, and street vendors). However, there are important distinctions between the characteristics of the two retailer types, as well as how consumers interact with them and the implications of those interactions on household food access and security.

Government and community public markets are a critical source of food for low-income households, but they also supply food to households across the socio-economic spectrum, with higher income

households using public markets to purchase fruits, vegetables, and wholegrain cereals (Khonje et al., 2020). Government public markets are managed by a government-appointed market manager and elected market committees through a top-down approach, and these markets serve as sites of revenue generation for the local government (Blekking et al. 2017). When formal institutions or organizations are unable to provide basic services to low-income households, then bottom-up mechanisms can be leveraged by communities to meet their food needs (Peyton et al. 2015). Community public markets, in which community members self-organize from the bottom-up to manage and ensure access to affordable food, are often less formally managed than government public markets but are nonetheless important (Davies et al. 2022), particularly in urban areas in Southern Africa where governance and infrastructure are often lacking (Maxwell 1999; Blekking et al. 2017). Both community and government public markets are often spatially distributed throughout an urban area, including in central business districts, urban residential areas, and along the transitional peri-urban fringe (Nodal Conseil, 2013). It is important to note that the delineation of public markets as government or community is useful for identifying which groups establish and manage markets; however, these broad definitions can obscure variability in market functionality. The functionality of public markets varies based on several criteria, including governance structures (Davies et al. 2022). For example, a government public market that grows and extends beyond its mandated location may be governed by formal processes in its core trading area, while the periphery may be more informally governed.

Economic theory states that as agricultural value chains in developing regions evolve or modernize, supermarkets become the main form of food retail and public markets begin to phase out as food systems transition from traditional to modern systems (Barrett et al. 2020). However, whether public markets in large African cities will phase out is not clear, because public markets continue to persist in urban African food systems (Hannah et al. 2022). The persistence of public markets is important for urban food security because economic gains in Africa are not keeping pace with urbanization rates, meaning urban poverty rates remain high (World Bank 2018). Low incomes limit the types and quantities of foods that consumers can purchase, thus influencing which retailers they interact with. As a result, research suggests that public markets remain viable food distributors within food retail environments through their ability to provide fresh, affordable, and culturally preferred foods (Hannah et al. 2022).

When compared to public markets, supermarkets are a relatively new component of urban food systems in Southern Africa. Starting in the early 1990s, supermarkets began to emerge more widely in developing nations, largely because of structural adjustment programs that emphasized market liberalization. Market liberalization enabled an investment environment that featured increases in foreign direct investment, increases in supply chain expertise, growth of competitive business practices, and development of agricultural production and food processing technologies in countries across Southern Africa (Barrett et al. 2020). To this day, policymakers often welcome the introduction and development of these changes to develop and modernize economies (UN Habitat, 2018), and these changes have led to integration of more global and regional value chains. At the same time, urbanization and consumer-related trends also contributed to rising demand for supermarkets. Higher rates of urbanization are typically associated with rising household incomes and consumers increasingly demand access to processed foods and foods that are high in animal products, as well as greater demand for non-nutritive attributes (e.g., food safety, appearance, convenience, storability, taste, and variety) (Reardon et al. 2021; Seto and Ramankutty 2016; Barrett et al. 2020). These changes benefit international and regional companies, like foreign-based supermarket chains, by allowing access to previously inaccessible markets. However, the increased role of private sector actors within the different parts of the food system can contribute to debates about sustainability, equitable livelihoods, and food system participation (Clapp, 2021). Some argue

that an increased role by private actors can improve efficiency and competitiveness, while others argue high rates of private investment can inhibit equitability and hinder food security (Rousseau et al. 2020). In Southern Africa, public-private agreements have enabled the development and growth of the supermarket sector. Supermarkets in the region are established and operated by private companies (das Nair 2018), but may also receive government incentives in terms of tax breaks or infrastructure development (e.g., power grid, transportation networks, water points and connections, among others) that food supply chains require (UN Habitat, 2018).

The development of the supermarket sector has done little to ameliorate widespread food insecurity in urban Southern Africa, because purchasing from supermarkets and other similar food retailers is largely associated with higher income (Wanyama et al. 2019; Berger and van Helvoirt 2018). Yet, supermarket development continues to be at the forefront of food retail development in many Southern African cities. Consumer groups from across the socio-economic spectrum purchase from supermarkets (Khonje & Qaim 2019), but supermarkets are not well-aligned with the food purchasing needs of low-income households. Low-income households require food to be sold in smaller, more affordable quantities than what is offered by supermarkets (Battersby & Watson 2018). Public markets, on the other hand, adapt to consumer needs by repackaging food into smaller quantities that consumers can afford to purchase on a more frequent basis, and more easily carry when using public transportation. Some vendors in public markets also offer credit to repeat customers and stay open longer than supermarkets or smaller grocers. The adaptability of public markets is one reason why low-income households tend to rely on public markets as a primary source of food, despite the presence of supermarkets (Hannah et al. 2022; Mwango et al., 2019; Frayne et al. 2010).

3. Study site and data

Zambia is a landlocked country in south-central Africa with a population of approximately 18 million people (World Bank, 2023). In 2018, slightly <8 million people (44% of the population) lived in urban areas of all sizes (United Nations 2019). In 2020, the capital city of Lusaka was home to approximately 2.2 million people (~12% of the country's population (CSO 2020)). By 2035, the city's population is projected to exceed 4.5 million people (CSO 2013). Multiple studies have used Lusaka as a case study for empirical food- and urban-related research. For example, for an evaluation of urban food security levels across Lusaka (Blekking et al. 2020); for a study of an urban food system's transformation (Abrahams, 2010; Khonje and Qaim 2019); for an assessment of urban spatial development patterns (Simwanda and Murayama, 2018); as part of a comparative study on policy responses toward informal traders (Resnick, 2019); or as the location of a regional assessment of a food system (FAO and the RUAF Foundation, 2019). The area of Lusaka is approximately 419 square kilometers, and the city contains 94 residential areas, with 66 designated as planned and 28 designated as unplanned, according to 2020 data provided by the Lusaka City Council's Geographic Information Systems Unit (Fig. 1) (LCC GIS Unit, 2020). In 2007, approximately 70% of the population of Lusaka lived in unplanned residential areas (UN Habitat, 2007). Planned residential areas in Lusaka are developed through government authorization, while unplanned residential areas are developed independently by residents (Simwanda et al. 2020).

This analysis uses a dataset of supermarket and public market locations and a dataset of population estimates. We developed the supermarket and public market datasets through a multistep geo-referencing procedure. First, we identified the general locations of supermarkets in Lusaka by using supermarket company websites (i.e., store finder features), shopping mall directories, and company Facebook pages (similar to the approach used by Berger & van Helvoirt, 2018). The dataset includes multinational supermarket chains (Pick n' Pay, Shoprite, Food Lover's Market, Game, SPAR, and Choppies) and local supermarket

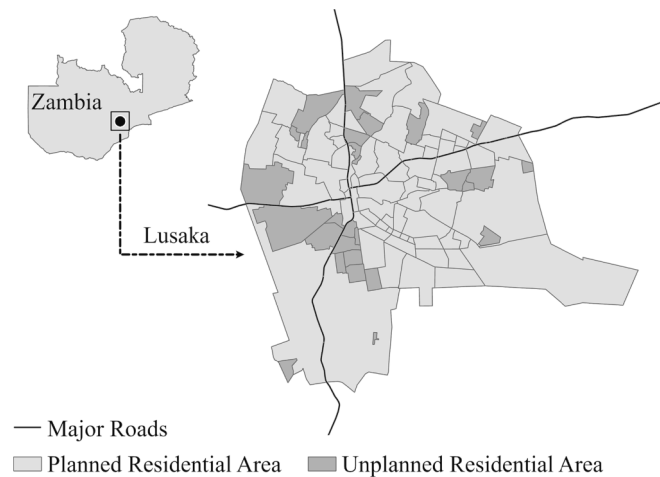


Fig. 1. Location of Lusaka, Zambia, and the distribution of unplanned (28) and planned (66) residential areas within the city in 2020 (94 residential areas in total).

chains (Melissa and Cheers). Khonje and Qaim (2019) use the same supermarket chains in their study of food consumption behaviors in Lusaka. The dataset of public markets is based on a list of markets provided by the Zambia National Marketers Credit Association (Zambia National Marketers Credit Association, 2020), and exact geographic coordinates (i.e. latitude and longitude) of public markets were found using Google Maps™. Second, once the locations of supermarkets and public markets were identified, we used Google Earth Pro™ imagery and worked backward from 2020 to 2004 using all available imagery (described in more detail below) to identify when the respective supermarkets and public markets were established. For public markets, we used a combination of completed buildings and the presence of cars as indicators of when the public market began operation. We use the combination of a completed building, a car park, and the presence of cars as indicators for when the supermarket was established. For instance, in a scene featuring the Cheers supermarket located in the Chalala residential areas from October 2017, a completed building is seen, but the parking lot is incomplete, and piles of earthen material are visible. The next available scene is from March 2018 and passenger cars appear in the completed car park. This location was then indicated as established in 2018. We do not differentiate intra annual images in our analysis. For instance, images used from January 2019 or April 2019 are both indicated as 2019.

Imagery obtained from Google Earth Pro is not readily available or reliable prior to 2004. In these instances, nine supermarkets and 51 public markets were indicated as having been established in 2004. The first supermarket opened in Zambia just nine years earlier in 1995 (Badal, 2021). Google Earth Pro™ images for Lusaka are of poor quality for 2007 and 2008 due to blurriness. Two public markets and two supermarkets were established in 2007 or 2008, meaning we cannot be sure which of those two years they were finally completed. For these locations, we indicated the year of construction as 2009, the next year that clear images were available. For the other years of our study period, Google Earth Pro™ imagery of Lusaka is cloud free and multiple scenes exist for each year.

Supermarkets can differ from one another in substantive ways, such as their market competitiveness (das Nair, 2018), food product quality (Battersby & Peyton, 2014), and floor space (Khonje & Qaim, 2019). We do not discriminate between different forms of supermarkets (i.e., lower-end supermarkets versus higher-end supermarkets). Since the onset of the Covid-19 pandemic in 2020, the supermarket chain SPAR shuttered all of its stores in the country in late 2020, except for four independent stores (Chanda, 2020), while Food Lover's Market temporarily closed its stores in Lusaka in 2021 (Researcher personal communication, 25

February 2022). We still include both supermarket companies in our analysis because the analysis concluded in 2020. For public markets, we differentiate between the community and government public markets to understand the differences between bottom-up (community) and top-down (government) market development trends within the food retail environment of Lusaka. Community and government public markets are categorized using a list of government-managed markets created by the International Growth Centre (International Growth Centre, 2020).

For population estimates, we use the unconstrained individual countries 2000–2020 UN adjusted (100 m resolution) dataset made available by World Pop (World Pop, 2022). This dataset estimates the total number of people per 100 m grid-cell, with the estimates adjusted to match corresponding population estimates from the United Nations (United Nations, 2019). World Pop datasets are estimated to be 60% accurate (Bai et al., 2018). Annual residential area population density values were created by clipping the gridded values of the annual World Population dataset using the boundaries of the 94 designated residential areas. We differentiate between planned and unplanned forms of residential area development, because households located in unplanned residential areas often contend with higher rates of poverty, poorer access to public services and infrastructure, and higher food insecurity rates than households in planned areas (Frayne & McCordic, 2015; Simwanda et al., 2020). We then calculated the sum of all cell values within the respective planned and unplanned residential areas using zonal statistics. The annual residential area population density estimates were then divided by the area of the respective residential area to provide an annual population density estimate for each residential area from 2004 to 2020.

4. Statistical Analysis

We first compare the number of supermarkets developed to that of community (bottom-up) public markets and government (top-down) public markets in Lusaka from 2004 to 2020. We then calculated the mean population density change for unplanned, planned, and combined (i.e., planned and unplanned) residential areas in Lusaka for the study period. The community and government public market and supermarket locations were then overlaid on top of the mapped population densities and examined for five years: 2004, 2008, 2012, 2016, and 2020. We use these five years to evenly divide the period of analysis for visualizing our findings. This was done to understand the patterns of development of the different types of retailers across changing residential areas throughout the study period.

To identify the spatial distribution and temporal rate of supermarket and public market development in Lusaka, we used Kulldorff's space–time scan statistical analysis conducted in SaTScan version 9.6 (Information Management Services, Boston, MA, USA) (Kulldorff 1997). SaTScan is a form of cluster analysis, which is used to identify where and when statistically significant high and low densities of occurrence are occurring across space and time. SaTScan was first developed to detect disease outbreaks and other health outcomes (Kulldorff 1997). To our knowledge, this is the first study to evaluate spatial–temporal food retail development trends in Africa using SaTScan. We use this approach for two reasons. First, food security and food systems are both spatially and temporally nuanced (Widener and Shannon, 2014), so understanding how the food system is changing across space and time requires such an analysis. Second, the on-going transformation of food systems in Southern Africa requires sub-city scale spatial–temporal assessment to improve our understanding of how transformations impact the food security and lived experience of urban residents.

We use the Bernoulli model within SaTScan to conduct our spatial–temporal analysis. The Bernoulli model assumes that cases (supermarkets) have the same geographical distribution throughout the study period as the controls (public markets). We use public markets as the control points because they serve as the largest historical direct-to-consumer and wholesale retailer, and most locations were established

long before supermarkets. The Bernoulli model has three assumptions. First, that each trial will be a success or failure. Second, the probability of success or failure of each trial is constant. Third, each trial is independent, and the likelihood of each outcome does not change from trial to trial. The simulations are conducted by imposing circular windows of different radii (the spatial unit) over each control and case location and over various time periods (i.e., one year, two years, etc.) (Kulldorff, 2018). This is done to identify potential localized clusters without incorporating prior assumptions about their size or location, thus the assumptions required by the Bernoulli model are maintained. For statistical inference, 999 Monte Carlo replications were performed using a circular scan with a maximum spatial cluster size of 50% of the total population. We also used a minimum temporal cluster of 1 year, with the maximum set to the entire study period (17 years). The null hypothesis of spatial randomness was rejected when the simulated p-value 0.05. When the number of cases (supermarket) is significantly higher or lower than the expected number compared to the controls (public market) within the scan window, a statistically significant “cluster” of supermarkets is identified. We set the model to not allow geographical overlap of similar high or low outcomes, meaning that a cluster of high supermarket development will be reported if it does not overlap with a more likely and previously reported high cluster and if it has the highest maximum likelihood. Importantly, an area of low supermarket development does not mean that an area is completely without supermarkets, rather the analysis indicates that the supermarkets within that area are not significantly clustered in space or time because the observed number of supermarkets in that area are not significantly different than the expected number. We then overlaid these clusters on the population density dataset to identify areas where supermarket development is most concentrated or absent in relation to changes in the population density for individual residential areas from 2004 to 2020. We do not correct for edge effects in this analysis because there are no supermarkets or public markets within 5 km of the city’s limits.

5. Results

The total number of public markets in Lusaka increased from 52 in 2004 to 65 in 2020 (Fig. 2). When disaggregated, community public markets increased by 10 markets (~25%) from 27 in 2004 to 37 in 2020, while the number of government public markets increased by 3 (12%), from 25 to 28. Using available population estimates for 2011 and 2020 (CSO 2013), we find a decrease in the number of public markets per capita from approximately one public market per 30,934 people in 2011

to approximately one public market per 42,015 people in 2020. The number of supermarkets increased from 9 in 2004 to 54 in 2020, an increase of 500%. This represents an increase in the number of supermarkets per capita from approximately one supermarket per 145,153 people in 2011 to approximately one supermarket per 50,574 people in 2020. Prior to 2012, the rate of supermarket development was relatively stable with between 0 and 2 supermarkets developed per year; however, beginning in 2012 more than 5 supermarkets per year were developed annually. The year with the greatest increase in the number of supermarkets occurred in 2016, when 9 supermarkets were developed.

The nine supermarkets that existed in Lusaka in 2004 were largely clustered around the two primary roads that pass through the city: the Great East Road that extends to Malawi, and the Great North Road that connects Cairo, Egypt to Cape Town, South Africa (Fig. 3). The Great North Road differentiates the eastern and western halves of Lusaka and is the most used road in the city. Expansion of the supermarket sector first radiated outward from the city center toward the less-densely populated, typically higher income southeast region of the city between 2004 and 2008. In 2004, population density for all residential areas was below 6,000 people per square kilometer. From 2008 to 2012, supermarket expansion further increased near to the city center and toward the northern, more densely populated areas of the city, which is bisected by the Great North Road. From 2012 to 2016, supermarkets then began expanding along the southern extent of the Great North Road and toward the eastern edge of Lusaka. These areas have low to middle density populations compared to the rest of the city. Finally, from 2016 to 2020, supermarket development strengthened in the eastern and southeastern areas of the city, where population densities are mixed. By 2020, the average population density in Lusaka was nearly 8,000 people per square kilometer, and some residential areas had exceeded more than 12,000 people per square kilometer. During the study period, the more densely populated residential areas in the western half of Lusaka experienced little increase in the number of supermarkets developed, but this region is where 5 of the 11 (45%) newly established community public markets were located.

From 2004 to 2020 each of the 94 residential areas in Lusaka experienced a population density increase of at least 150% (Fig. 4). In general, the western half of Lusaka experienced the greatest rate of population density change. Many of the residential areas grew by over 300%, with the most significant changes occurring in Linda (381%), Chibolya (365%), and Garden Park (349%) residential areas. All three of these locations are unplanned residential areas in the western half of the city. The lowest percentage population density change occurred in the

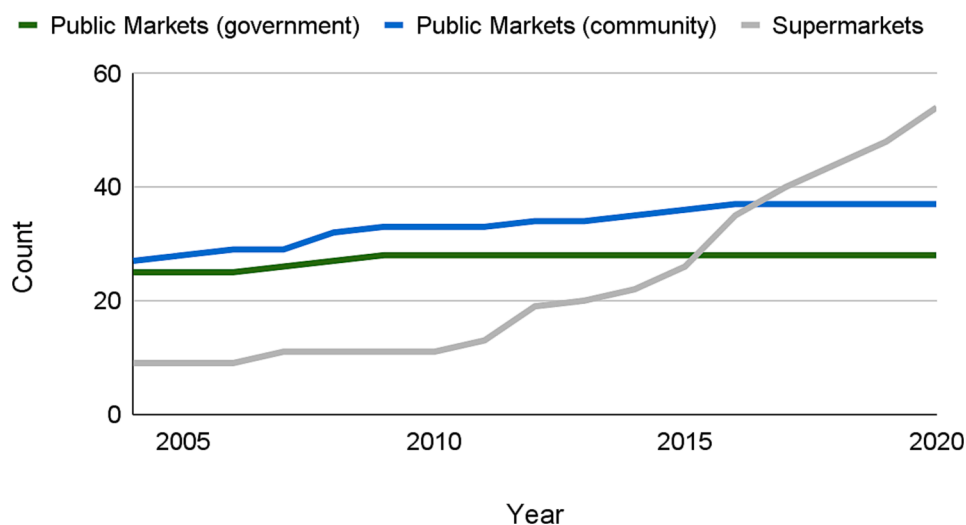


Fig. 2. The count of government public markets, community public markets, and supermarkets in Lusaka, Zambia, from 2004 to 2020. The development of public markets is relatively low from 2004 to 2020 in comparison to supermarkets, which have increased by 500%.

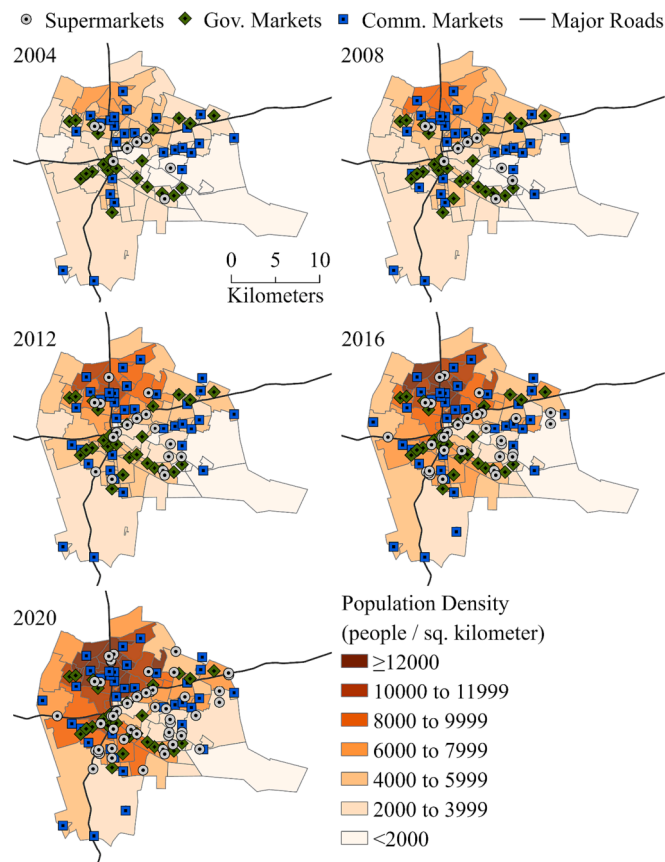


Fig. 3. The development of supermarkets, government public markets, and community public markets overlaid on population density estimates for the 94 residential areas in Lusaka from 2004 to 2020. Supermarket development occurred more often in the eastern, less-densely populated half of Lusaka, while the more densely populated western half of the city has fewer supermarkets, government public markets, and community public markets overall.

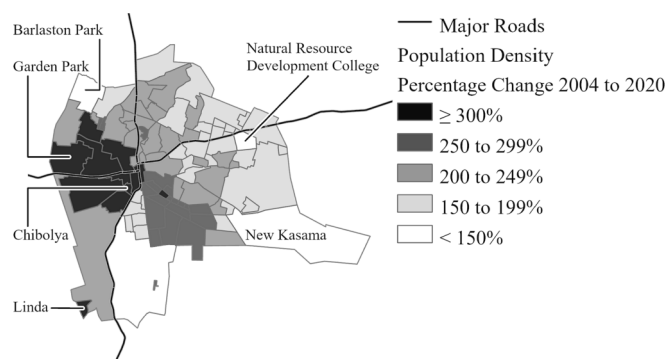


Fig. 4. The percentage change in population density of the 94 residential areas in Lusaka from 2004 to 2020. White colored areas on the map represent residential areas with population density percentage change of less than or equal to 150%, while black colored areas on the map represent population density percentage change of over 300%.

Natural Resource Development College (128%) and New Kasama (145%) residential areas in the eastern half of the city, and in Barlaston Park (141%) residential area in the northwest region of Lusaka. Each of these three residential areas are designated as planned.

Both unplanned and planned residential areas, on average, doubled in their density during the 17-year study period (Fig. 5). In 2004, the residential area with the highest population density in the city was the

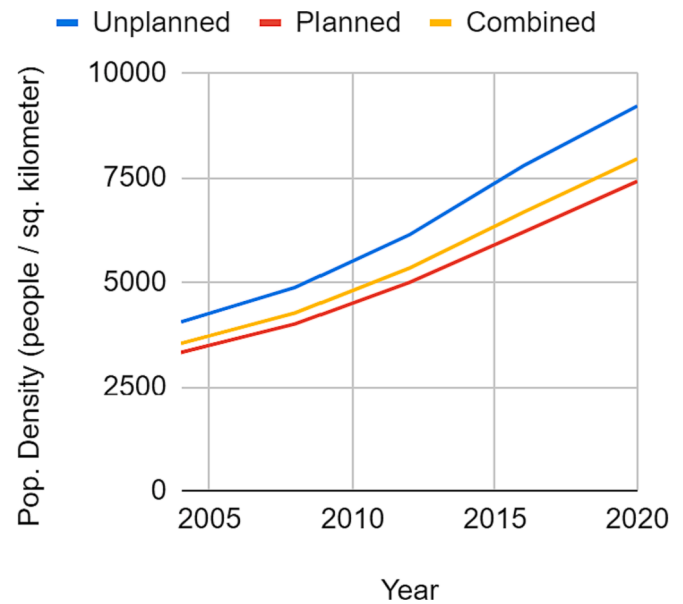


Fig. 5. Mean population density growth from 2004 to 2020 for unplanned, planned, and combined (both planned and unplanned) residential areas in Lusaka, Zambia.

unplanned residential area of Chunga, located in northwest Lusaka, with 7,777 people per square kilometer, while the lowest was the planned residential area of New Kasama (southeast Lusaka) with 364 people per square kilometer. In 2004, the 28 unplanned residential areas had, on average, 4,057 people per square kilometer, while the 66 planned residential areas had an average of 3,327 people per square kilometer. In 2020, New Kasama had the lowest population density with 529 people per square kilometer, while the planned residential area of Matero East (northwest Lusaka) had the highest population density with 16,897 people per square kilometer. On average, unplanned residential areas in 2020 had a population density of 9,224 people per square kilometer while planned residential areas had a population density of 7,426 people per square kilometer.

SaTScan results indicate statistically significant ($p < 0.05$) areas of low supermarket development in Lusaka between 2004 and 2014, and areas of statistically significant ($p < 0.05$) high supermarket development between 2013 and 2020 (Fig. 6). Statistically significant areas of low supermarket development represent areas that had fewer than expected supermarkets compared to public markets given the Bernoulli distribution of the spatial-temporal data (i.e., locations of supermarkets and public markets each year). From 2004 to 2014, four areas of statistically significant ($p < 0.05$) low supermarket development existed in Lusaka. The largest area of low supermarket development had a radius of 7.04 km, covered much of the eastern edge of the city, and occurred from 2006 to 2013. Substantively, this indicates that there were fewer than expected supermarkets as compared to public markets within a large portion of the city from 2004 to 2014, and the large size of this cluster represents a dispersion of public markets (the control) compared to supermarkets (the cases) over a large area. The smallest area had a radius of 1.65 km and occurred near the city center between 2004 and 2011. The end date for areas of low supermarket development corresponds with when supermarkets in those areas were established, thus the spatial pattern of supermarket and public market distribution changed from clustered to random (i.e., moved from significant to insignificant). Overall, these clusters indicate a period in which supermarkets were not significant actors in the spatial coverage of the food retail environment in Lusaka. During this period, public markets were significantly more prevalent across the city's food retail landscape.

The spatial configuration of Lusaka's food retail environment began to change in 2012 with the development of more supermarkets. Our

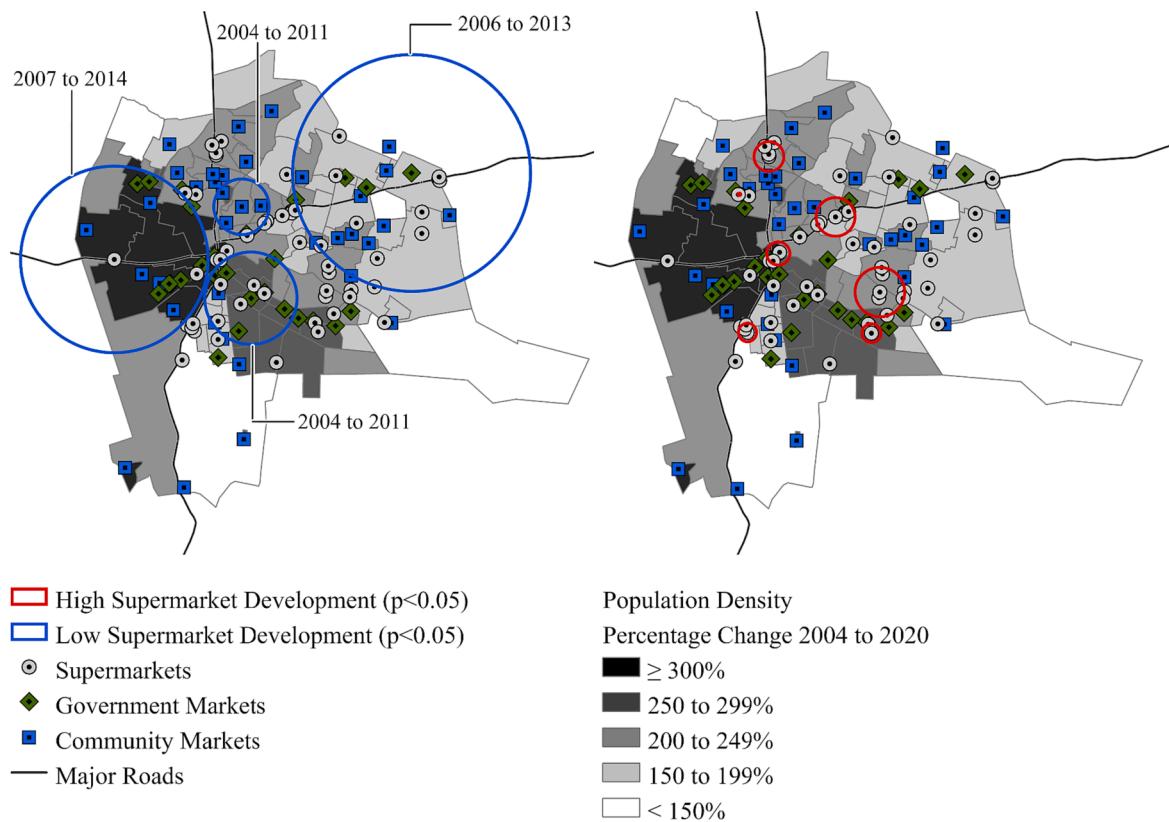


Fig. 6. SaTScan results indicate four statistically significant areas ($p < 0.05$) of low supermarket development in comparison to the establishment of public markets from 2004 to 2014 in Lusaka, Zambia. From 2013 to 2020, seven more dense areas of high supermarket development occurred. The supermarkets and public markets indicated on this map are for the year 2020.

results do not indicate why the uptick of supermarkets began in 2012; however, the increase may be the result of an improving economy, a new national government, and Lusaka's population growth. Starting in the late 2000's, Zambia's economy began to improve considerably. In 2013, GDP per capita reached its highest rate between 2004 and 2020 (World Bank, 2023), which increased consumer purchasing power. Around the same time, the Patriotic Front political party won the national elections, in part by promoting a policy platform that sought to increase foreign direct investments, like international supermarkets. Third, as previously outlined, Lusaka's population considerably grew during this period. These three aspects together led to a construction boom in the city that began around 2012 (Chungu and Dittgen 2021), which the rapid growth of the supermarket sector has contributed to. Our results support this hypothesis, as we find that seven areas of statistically significant ($p < 0.05$) high supermarket development occurred from 2013 to 2020 (map at right in Fig. 6) - during the period of high GDP per capita and within the context of policy creation aimed at increasing foreign direct investment. In general, the seven statistically significant areas of high supermarket development are smaller and located in either lower density and higher income areas, near main roads, or both. These findings are similar to those of Berger and Helvoirt (2018) and Peyton et al. (2015), which found supermarket presence is low in urban areas with high poverty rates. The small sizes of the clusters indicate a higher density of supermarkets for that area, while the proximity to roads may increase physical accessibility to supermarkets. Importantly, these clusters also lack government and community public markets, and the development of supermarkets in these areas may improve physical access to food. The largest area of high supermarket development had a radius of 1.15 km, while the smallest area had a radius of 0.07 km. Four of these clusters are located along the Great North and Great East roads in Lusaka, where public transportation routes readily exist. All but one of these seven areas are in residential areas where population densities

increased between 150 and 249% from 2004 to 2020.

6. Discussion

In this discussion we first contextualize the development of supermarkets and public markets in Lusaka from 2004 to 2020. We then discuss why city planners should consider the interconnectedness of transportation, water, energy, and food systems to minimize urban inequalities. However, this consideration may be a challenge because many policymakers and planning units favor some forms of urban food retail while delegitimizing others. We then discuss how the western half of Lusaka is significantly less developed in terms of food retailers, according to our spatial-temporal analysis. This area may be well suited for public markets, but government efforts are stagnant, while community public market development may be limited by the availability of resources. Finally, we discuss how public-private partnerships may help to limit food retail environment vulnerabilities and inequalities.

The distribution of supermarkets across space and time in Lusaka, Zambia, suggests that this sector of the food retail environment is developing unevenly. Our results do not indicate that supermarkets are replacing public markets - the total number of public markets increased from 2004 to 2020, but we find greater supermarket development in lower density, higher income areas of the city. In higher density, lower income areas of Lusaka, supermarket development is low or non-existent. This finding is similar to work by Peyton et al. (2015); however, our results also show that during the same period and in the same areas, public market development does not significantly increase, despite the role that public markets have in improving food access, particularly for low-income households. The structural factors (e.g., uneven infrastructure development, poor service provision, and widening income gaps) that contribute to higher or lower rates of poverty in particular urban areas likely contribute to the spatial

distribution of supermarkets. It is unclear whether the growth of the supermarket sector will continue at the pace and scale observed in our results due to the unique urban development trends in Southern Africa, notably that urbanization rates are outpacing economic gains and urban poverty rates remain high (World Bank, 2018). Global development trends and crises such as the COVID-19 pandemic, international conflicts and food price increases, and climate change impacts each illustrate the need to develop food retail environments that can withstand multi-scalar shocks while meeting the needs of urban residents from across disparate urban areas and the socio-economic spectrum. Uneven spatial and temporal development of supermarkets and public markets may substantially limit the ability of food retail environments in large Southern African cities to improve food security in the future.

Many structural inequalities in Southern African cities are linked to long-standing colonial legacies. During the colonial era, investments in urban areas were generally targeted toward European minorities, thus excluding whole areas, and marginalizing certain groups (Fox, 2014). Policymakers to this day continue to marginalize areas and groups by following colonial conceptualizations of the modern city (Kiaka et al. 2021). Contemporary policymakers often view supermarkets as clean and modern, discounting other forms of food retail (e.g., public market and street vendors) as undesirable and backwards (Battersby and Watson 2018). Modernization efforts related to food systems tend to favor food system development centered around supermarkets and removal of street vendors, shops, and public markets that do not adhere to government oversight (Battersby, 2017; Haysom and Fusieni 2019; Resnick, 2019; Kiaka et al. 2021). Limited development of supermarkets in some residential areas and a parallel reduction in government involvement in public market development and management can leave some areas without physical access to necessary food retailers, thus further marginalizing urban households that contend with socio-economic exclusion. Policy and planning efforts aimed at ensuring food systems can meet the needs of different socio-economic groups located in different areas of a city will require deliberate consideration of other key infrastructure and services like transportation, water, land, energy, healthcare facilities, and social support mechanisms. This is because food systems are not siloed, rather they are closely interlinked with other urban systems that directly or indirectly affect how people access food.

Our SaTScan results indicate significant areas of low supermarket development in the early part of the study period, with the second largest area of low supermarket development located in the western half of Lusaka where many residential areas are unplanned and more densely populated. According to Simwanda et al. (2020), households in unplanned settlements in Lusaka often contend with high levels of poverty, lack necessary infrastructure like water, sanitation, and electricity, and have limited access to education and health services. The largest area of low supermarket development (indicating dispersed public markets) is located along the eastern periphery of Lusaka and overlays a mix of planned and unplanned areas. No areas of significant low supermarket development exist for the second half of our study period, yet our maps indicate areas with few to no supermarkets in areas with substantially high population density increases (e.g., southern, western, and north-eastern Lusaka). Our results suggest that policymakers and private sector supermarkets are not facilitating the development of the food retail environment in more dense, lower income areas, which has important implications for the food retail environment and urban household food security. A potential policy intervention to remedy the uneven development of the local food environment would be the development of public markets; however, our results show limited development of government public markets during the study period.

The limited number (3) of government public markets developed from 2004 to 2020 suggests that the government was slow to develop new markets in Lusaka, despite the city's increasing population density rates and associated increase in demand for food. Our data do not indicate why the government developed so few markets during the 17-

year period, but one reason may be related to a reconsideration of the role of government in market development and management that is ongoing in other African countries. For instance, food retailer policy discussions in Morocco are split between those that want to privatize public markets to improve efficiency and competitiveness, and those that favor an increased role of the government to develop markets and improve equitability (Rousseau et al. 2020). Another reason may be related to decentralization and service provision. Resnick (2014) states that as many African governments become decentralized, local governments are tasked with providing more services to constituents, yet high levels of poverty, administrative hurdles, and limited resources can hamper service provision, such as public market investments. If this is the case, policymakers may lack necessary resources to invest in the food retail system, while private food retailers like supermarkets may gain an increasing role in food retail system investment through their ability to leverage private capital. However, emphasizing the role of private food retailers without considering how the structural roots of urban inequalities can limit food security can lead to unequal food outcomes. For example, corporate retailers, like supermarkets, try to maximize demand and thus profits by establishing stores near major roads and intersections (Haynes and Fotheringham 1985); however, transportation access is not always equitably distributed across cities. Shannon (2021) states that corporate food retailers and partnering financial institutions often enhance unequal food access and exacerbate existing urban inequalities by specifically avoiding investment in and development of stores near certain areas or groups of people. This form of economically liberalized development can exacerbate existing inequalities, enhance potential food system vulnerabilities, and decrease food system resilience. However, as previous work by (Haysom 2021) states, urban planners in Southern Africa often do not believe, and at times are not empowered, to address food systems and food security. Failing to incorporate a broader understanding and approach to food systems, including the spatial distribution of different food retailers and systemic underpinnings that marginalize certain groups, in urban development and food security policies may further hinder food security in the city.

In the absence of government public market development, community public markets can play an important role in improving food access. Our results suggest that urban residents have self-organized to establish and govern markets through bottom-up efforts to serve the needs of their communities. A bottom-up approach to market development is promising in that it suggests community organization and collective action approaches to solving a problem, but this finding should be contextualized. Community public markets often lack basic infrastructure and services like sanitation, waste removal, and electricity, which can hinder the ability of the market to operate effectively (Davies et al., 2022). Community public markets may also suffer from a lack of political and legal support. Governments throughout Southern Africa have harassed and removed vendors in community public markets (Haysom & Fusieni, 2019), turned a blind eye to markets being used as locations of political contestation (Resnick, 2022), and invoked public health policies to shut down and remove illicit markets (i.e., those not formally registered with the government) (Kiaka et al., 2021). For instance, during a cholera outbreak in Zambia in 2017, local government officials destroyed the stands of community public market vendors throughout high-density areas in western Lusaka to eradicate possible sites of disease transmission (Resnick, 2018). For community public markets, failure to provide essential services like sanitation and water access can increase their vulnerability to environmental and food safety hazards, thus enhancing their chances of being targeted by public health authorities. The removal of market vendors or shuttering of facilities can exacerbate existing food insecurity challenges by limiting access to affordable food and impacting the livelihoods of sellers, many of whom are also residents within the community.

Development of a food retail environment consisting of food retailers that source through long, complex supply chains are susceptible to global shocks through environmental, political, social, and economic

pathways, and these susceptibilities can leave residents in African cities more vulnerable and exacerbate existing urban inequalities (Moseley and Battersby 2020; Blekking et al., 2022). For example, the COVID-19 pandemic and associated economic slowdown pushed the supermarket chains SPAR and Food Lover's Market to exit Zambia or temporarily close their stores across the country. The recent COVID-19 supply chain shocks also illustrated how food retailers rely on varied and complex supply chains, and that shocks can lead to both shortages and overabundance (Barrett et al. 2020). Developing urban food environments where retailers of different sizes source food from aggregators, distributors, and producers across varied supply chains, can help to increase equitable economic and spatial access to food, while also improving resilience to shocks near and far. One approach could involve a public-private partnership that enhances resilience and equitability through the development of different forms of food retailers that source from variable supply chains. For example, in Nanjing, China, urban planning policies require private developers to create wet markets in newly developed residential areas and also mandates that supermarkets have at least 20% of their floor space dedicated to fresh produce (Zhong et al., 2019).

7. Conclusion

Cities are rapidly growing in Southern Africa. At the same time, their urban food systems are transforming, although this transformation is not always even across space or time. We find that from 2004 to 2020, supermarkets developed over twenty times faster in Lusaka than public markets, and that supermarkets are typically established in less dense, higher income areas of the city, even though these areas are not where the greatest rates of population growth (and thus food demand) are occurring. We find statistically significant clusters of high and low supermarket development across different areas of Lusaka, and these clusters illustrate the uneven growth of the food retail environment over space and time. The development of the supermarket sector is not inherently detrimental to urban food systems or the food retail environment if other food retailers are also supported through planning and policy efforts. Without concomitant development and support of public markets and other food retailers, it is unlikely that food access or food security in Lusaka will improve over time and space. A pressing policy challenge for the development of food retail environments in large Southern African cities is incentivizing a development trajectory that lessens existing food-related inequalities as urban areas expand and populations grow. This will require making and implementing policies and investments that promote a food retail environment capable of meeting the demands of all urban consumer groups across the socio-economic spectrum, while reducing potential food system shocks. Urban Southern African households and the residential areas they reside in are socially and economically diverse and require a diverse range of public and private food retailers of varying sizes, that operate in variable ways, and are evenly distributed across the urban landscape. Uneven development of the food retail environment, both in terms of economic and spatial access, hinders equitable food security and food system resilience in rapidly growing urban areas. By planning for and developing policies that promote both supermarkets, public markets, and other necessary food retailers, urban policymakers will begin to make inroads on increasing food access, food security, resilience, and equitability of food systems in Southern Africa's growing cities.

CRedit authorship contribution statement

Jordan Blekking: Conceptualization, Data curation, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Kurt Waldman:** Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition. **Julia Davies:** Conceptualization, Methodology, Investigation, Writing – original draft, Writing –

review & editing. **Mboyonga Kaputula:** Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Erik Nelson:** Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Scott M. Robeson:** Methodology, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoforum.2023.103893>.

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